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*Terms and Conditions apply**The future of the 'future' of mobile networks**

In telecommunication circles, the evolution to 4G is a topic that has garnered significant interest from operators and network equipment manufacturers alike. With LTE being touted as the future of mobile broadband communication, communication test equipment companies will play a pivotal role in the development, support and optimisation of LTE. The LTE test equipment market is highly competitive with several companies vying for the position of the leading technology provider with solutions that incorporate the latest standards and technologies. As a result, an in-depth understanding of the trends impacting the LTE market is imperative to maintain a competitive edge in this market.

Anil explains, "As LTE is a technology is a fully packet-switched network, focused on high-speed data services, more emphasis on ensuring QoS and key performance indicators is a must while testing. Rohde & Schwarz has developed the PQA test system, exactly for this purpose. PQA ensures that all key performance indicators of the device are fully tested for optimal performance under field conditions."

With only about 9 per cent fixed-broadband penetration of households, mobile broadband is well-positioned to become the main means of Internet access for many Indians. This has already started with 3G, and LTE—both TDD and FDD—will further enhance access to the Internet over the coming years.

For equipment vendors and operators, growth in mobile broadband shifts the testing emphasis from simple compliance to 3GPP standards to more towards benchmarking and performance measurements. As network deployments accelerate, the numbers of handsets and devices 'supported' by LTE base station vendors is becoming a key marketing metric; however, some have struggled with real-world delivery of lab-measured metrics.

Hire adds, "In order to meet the capacity demands, there will be a

shift away from centrally-managed networks based on macro base stations to HetNets comprising a mix of home-based femtocells, small cells (micro- and picocells) for hotspot and in-building coverage, and traditional macrocells. Creating a heterogeneous network is an attractive approach to meet traffic demands and performance expectations, particularly in situations where traffic is concentrated in hotspots, or areas that cannot be suitably covered by the macro layer. A heterogeneous network combines big, traditional base stations with small cells. It will reduce carriers' reliance on large, macro base stations wherever they need coverage, and enable mobile operators and end customers alike to increase capacity where needed."

Yanamandra says, "The most and immediate push is towards making the handsets as well as the operators stabilise the interworking between the existing technologies and LTE. Apps development on the most prominent mobile OSs' like iOS, Android and Windows is at its peak and similarly the Network operators enhancing their value-added services utilising the data speeds in providing rich multimedia services like IPTV (which was never to be a reality earlier), LTE Direct, etc. This will be future of wireless."

Changing test landscape

The next generation of mobile devices will need to provide a mobile broadband experience that matches the continually increasing expectations of the various network operators. It will be necessary to test new LTE devices using a layer-by-layer approach, building up to an end-to-end test scenario that uses real-world signal conditions. The most difficult challenge will be to ensure that the performance is maintained throughout the cell, especially as the number of users in the cell, and hence the signal noise level, increases. ●

The author is a tech correspondent at EYF Bengaluru